

Demand for the Immediacy of Execution: Time *is* Money¹

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Abstract

Recent interest in the time-to-execution as a measure of market quality in order driven markets coincides with the emerging empirical research on this topic. We build on recent theoretical models of dynamic limit order book to construct an estimation procedure that tests the effect of the expected time-to-execution on the order aggressiveness, taking into account the simultaneous determination of the two variables in equilibrium, the selection bias, and the censoring of the time variable. Using very detailed data from the Tel Aviv Stock Exchange, we show that the reduction in the expected time-to-execution is an important determinant of order aggressiveness, and may account for over 50% of the spread. Moreover, we obtain qualitatively similar results for stocks and government bonds that are traded on the same platform, which suggests that immediacy considerations have a significant effect on the high frequency price dynamics regardless of the degree of information asymmetry. We also show that the expected time-to-execution explains the choice of order strategy better than the probability of execution, which was traditionally used in the literature. Finally, our results corroborate predictions of several theoretical models of order driven markets.

JEL classification: G10, G15 C35.

Key words: limit order book; order submission strategy; time to execution; simultaneous equations; limited dependent variables; discrete choice; selection bias.

1 Introduction

What drives the high-frequency price dynamics in an order-driven equity market? The vast bulk of theoretical and empirical literature tends to see information asymmetry as the main driver. This paper builds on recent theoretical models that show that traders' demand for immediacy of execution can be a powerful driver of high-frequency limit order book dynamics.¹ We present empirical evidence that the order submission strategies of traders in securities markets strongly depend on their desire to shorten the expected time to execution. We show that this behavior has a significant effect on the high frequency price dynamics and is consistent with theoretical predictions. We also show that this behavior is observed in equities, as well as in government bonds, which have a drastically lower degree of information asymmetry. This leads us to conclude that the demand for immediacy of execution has a significant effect (economically and statistically) on price dynamics, above and beyond the effect of information asymmetry.

The provision of immediacy in financial markets can take three basic forms. On the one extreme is the pure dealer market, where all traders get immediate execution and incur the monetary cost of immediacy by trading at the quoted prices. On the other extreme is the periodic call auction, where all traders get to trade at the same time and price, but no one gets immediacy, as all of them incur waiting costs. The limit order book is the only trading mechanism that permits traders to choose between the two types of costs. Impatient traders² are likely to demand immediate execution

¹Immediacy is one aspect of liquidity, which is different from the monetary transaction costs that are usually studied. Information-based and immediacy-based channels are not mutually exclusive.

²Traditionally, informed traders were assumed to be impatient (e.g., Glosten, 1994), however some models allow them to be patient as well (e.g. Kaniel and Liu 2006). Here we use the terminology of Foucault et al. (2005) and Rosu (2006) to distinguish between

and trade at the quoted price against the limit orders in the book, while patient traders may prefer to wait, post a limit order with their desired price, and offer an option of immediate execution at that price to future market orders.³ The ability to choose the level of execution immediacy must be appealing to traders in equity markets, since order-driven markets in various forms came to dominate equity trading around the world.

In the context of a limit-order book, the choice between various order submission strategies is not arbitrary, but varies by trader type. Keim and Madhavan (1995) show that "value traders", who rebalance their portfolio due to long-term considerations, use less aggressive order strategies. Managers of index funds and technical traders, who have relatively short-term considerations, tend to use price-aggressive order strategies. Barclay, Hendershott and Jones (2006) show that arbitrageurs trade practically at any price, as long as the timing of the trade is right; making them arguably the most impatient traders. The differing valuations of immediacy across trader types provide the necessary conditions for the trade-off between price aggressiveness and the expected time-to-execution that we test in this paper.

Until the mid-1990's the theoretical literature had mostly focused on two types of trading mechanisms: dealer markets and call auctions, where the demand for immediacy does not play a major role, thus the emphasis is on price discovery. Recent theoretical developments focus more on features that are unique to the limit order book. In dynamic models such as Foucault, Kadan and Kandel (2005) and Rosu (2006), the authors assume away information asymmetry, and let the order submission decision be driven by the fact that an aggressive order reduces the expected execution time (traders patient and impatient liquidity traders).

³Today, most markets combine elements of more than one basic form, but the identification of these basic features is helpful for the discussion of execution immediacy.

value immediacy). While price priority rules mechanically determine the effect of the submitted order price on its expected execution time,⁴ the insight in these papers is that in equilibrium, there is a feedback effect as well: the expected execution time also affects the choice of order strategy. These theoretical models have a clear and novel empirical implication: if the demand for immediacy is indeed important to traders, an increase in the expected time-to-execution should increase the price aggressiveness of submitted orders. Our paper is the first to test these predictions in a context of an equilibrium model. Our empirical findings strongly support the predictions of these models. In fact, we show that many variables used in the empirical literature to explain order aggressiveness directly, actually affect it indirectly through the expected time-to-execution channel.

We use data from the Tel Aviv Stock Exchange (TASE) in 2000, when it was still a pure limit order market without intermediaries or hidden orders. These features make it a clean setting to test the implications of theoretical models of a pure limit order book. TASE made all its data files: orders, cancellations and trades with cross-references to orders, available to us. Thus, we are able to avoid problems arising from using small-size samples, aggregate data, and trade direction inferences from prices. A potential concern is that the TASE is less liquid than the US or many European markets. To address this concern, we test whether liquidity differences systematically affect the dynamics of order submission by replicating Lo et al. (2002) analyses of time-to-execution for our sample of stocks, and obtain very similar results.⁵ This gives us confidence that the insights obtained from our study extend to other exchanges and securities.

We postulate a system of simultaneous equations to address the endo-

⁴The price priority rule implies that aggressive orders are executed first.

⁵Results are available from the authors upon request.

generality of the two variables of interest: the expected time-to-execution and the price-aggressiveness of an order. This way we account for selection bias, and for the fact that we only observe the time-to-execution of the chosen strategies, but not of the counterfactuals. In addition, given the frequent cancellations of orders, we employ an econometric methodology that accounts for censored observations and we manage to avoid the downward bias of the lifetime estimates.

Our main result is that for traders "time is money" - they are willing to submit a more aggressive order, losing on price dimension, if it means a bigger reduction in the expected time-to-execution. Notice that this is not a mechanical effect of the aggressive order on time, but the reverse, as predicted by the equilibrium models described above. It is not a proxy for information asymmetry, or for the probability of execution. This is a robust phenomenon that calls for more attention to the demand for immediacy as a determinant of high-frequency price dynamics. As an illustration of the economic magnitude of this effect, we show that under average market conditions our estimates suggest that as much as 56% of the bid-ask spread of a liquid stock (33% for a liquid government bond) can be seen as a compensation for the time-to-execution of an average limit order.

Since time-to-execution plays the key role in our paper, we would like to clarify its role in our framework. It is clearly not the time value of money, as we are interested in very short time intervals. One interpretation is that traders intrinsically care about the immediacy of execution of their orders, making the time itself a primitive of the model. Traders' impatience may arise due to economic reasons such as arbitrage, agency trades, time constraints, trader evaluation, or due to traders feeling at ease only when their order has been executed. Another interpretation, is that the time to

execution is a proxy for unobservable primitive factors that traders may care about.

While we use the expected time to execution, the literature has focused on an alternative, yet related concept: the probability of non-execution that is widely assumed to drive order submissions.⁶ It is an empirical question whether the probability of execution is a “primitive factor”, a proxy for the same unobservable primitives, or just a proxy for time-to-execution. Let us compare the two: the probability of execution cannot be defined without specifying a time frame, say execution within an hour. If the trader cares only about the probability of execution, this implies that his objective is a step-function with respect to time (see e.g. Admati and Pfleiderer, 1988).⁷ A more reasonable assumption is that trader’s impatience is monotonically increasing in the expected time-to-execution. We use a methodology that allows us to estimate the entire distribution of time-to-execution, which clearly contains more information than the expected probability of execution within a pre-specified time frame. The extant literature chooses the expected probability of execution, thus implicitly assuming that the additional information contained in the expected time-to-execution is of no use to traders. This claim is not supported by the data: the results show that time-to-execution is indeed a much stronger determinant of order aggres-

⁶We refer to many papers below, but the closest to ours in this context is Hollifield et al. (2003).

⁷Extending the expected execution time from 10 to 3,599 seconds is costless for the trader, but an additional extension of 2 seconds is very costly. After that, any extension is again costless. Clearly for an arbitrageur the difference between 2 seconds and 3 minutes is very important; similarly a broker that routinely executes trades a day later (as opposed an hour later) will not get much business. One can also see the demand for speed in the market. For example, trading volume in the QQQQ ETF migrated from the original AMEX to Island ECN because of differences in milliseconds in execution (Wall Street & Technology 2002); Also, the recent merger between the NYSE and ARCA was said to be driven among other things, to help the NYSE compete in a market that emphasizes execution speed.

siveness than the expected probability of execution, which is consistent with arguments presented by Heckman and Singer (1984).

We must also point out, that information asymmetry considerations that dominate the microstructure literature are not explicitly present in our framework. While adverse selection is clearly an important factor in price dynamics,⁸ we claim that it does not drive our findings. We conduct our empirical analyses separately on a sample of stocks and a sample of government bonds, which are traded on the same TASE platform. Using TASE data puts us in a unique position to test the importance of immediacy under different regimes of information asymmetry, since government bonds are significantly less prone to asymmetric information than stocks. We get qualitatively similar results for stocks and government bonds, which supports the claim that immediacy-based considerations have a significant effect on price dynamics regardless of the degree of information asymmetry.

Our results corroborate several empirical predictions of theoretical models of the limit order book: order aggressiveness declines in the proportion of impatient traders in the population, increases in the arrival rate of traders (as predicted by Foucault et al. (2005) and Rosu (2006)), and in the depth of the book (as predicted in Parlour (1998)). Our empirical results on the effects of volatility are in line with the finding of Hasbrouck and Saar (2002).

The remainder of the paper is organized as follows. Section 2 surveys the relevant empirical literature. Section 3 sets up the econometric model and discusses its intuition. Section 4 describes the data and the sample, and defines the variables of interest, including their predicted effects. Sections 5 presents the results and section 6 concludes. A detailed discussion of the econometric model, as well as robustness and identification issues is

⁸See O'Hara (2004).

delegated to appendices.

2 Literature Survey

Demsetz (1968) suggested that waiting costs should be an important determinant of the bid-ask spreads in an order-driven market. However, only recently the time-to-execution became the focus of theoretical and empirical studies.⁹ We discuss the relevant theory when making testable predictions below, while in this section we only focus on the empirical literature.

Engle and Russell (1998) and Engle (2000) develop and estimate autoregressive conditional duration models. Since detailed order data is rarely available for a large sample of stocks in US markets, these models concentrate on the duration between trades and on the characteristics of transaction prices. Handa and Schwartz (1996) extract the execution time of a hypothetical order from transaction time and price data, assuming that the order is executed as soon as a suitable transaction price is observed. Lo et al. (2002) show that this type of procedure may not produce a good proxy for the true time-to-execution. Battalio et al. (2002) use time-to-execution, along with other parameters, as a measure of the quality of limit order execution in an analysis of the Merrill Lynch decision on 1995 to stop routinely routing orders to the regional exchanges in favor of the NYSE. They endogenize the broker's routing decision and control for the order submission strategy, but they treat the execution time as an exogenous variable and do not account for cancellations. Hasbrouck and Saar (2002) use a dataset provided by Island ECN, where order submission time is available. They test whether asset price volatility affects the order price-aggressiveness and time-to-execution, but do it in separate and unrelated models.

⁹See Foucault et al. (2005) for an extensive discussion of the theoretical developments.

Lo et al. (2002) use a unique dataset from ITG, an institutional brokerage firm, where one can directly observe order submission information. They estimate an econometric model of time-to-execution, stating that it is an important dimension of market quality in limit order markets (see also SEC 1997). They use proxies for the aggressiveness of limit orders as exogenous variables predicting the time-to-execution, implying a one way causality. We have replicated their study using the TASE data and the results were remarkably similar, suggesting that we can generalize the insights from this paper to other markets. However, we go much further than Lo et al. (2002), making our econometric model consistent with an equilibrium approach, where both the expected time-to-execution and the price-aggressiveness of orders are determined endogenously.

Order submission strategies (or price-aggressiveness) are analyzed in many empirical papers, usually in the context of the explicit transaction costs associated with these strategies rather than their expected execution time. Harris and Hasbrouck (1996) evaluate performance measures for market and limit orders for various order sizes, and spreads. Griffiths et al. (2000) use order submission strategies to explain the price impact. Ellul et al. (2003) model the order submission strategy as an endogenous variable in a multinomial logit model. They use stock characteristics and market conditions as exogenous variables. Ellul, Shin and Tonks (2005) study the choice of order submission to different market venues of stocks traded in London. Rinaldo (2004) models the order submission strategy as the dependent variable in an ordered probit model. All the papers described above do not consider the effect of waiting costs on the order submission strategy. Presumably some of this effect is picked up by other variables.

The most closely related paper is Hollifield et al. (2003), who use a com-

peting risks approach to estimate the probability of execution and discrete choice to estimate the order submission strategy. They postulate a rich theoretical model, in which traders' strategies depend on their private values, a common value and market conditions. They follow the theoretical setting very closely and make a large number of structural assumptions to estimate model parameters. They do not address the price-time trade-off, while our goal is to test the existence of a trade off between price aggressiveness and time-to-execution and we are less interested in parameter values. Thus, we confine our analyses to the qualitative characterization of this trade-off, by making only the necessary few structural assumptions to correct for several econometric problems. The two papers complement each other.

We use a simplified version of the Hollifield et al. (2003) specification to illustrate the informational advantage of the expected time-to-execution over the expected probability-of-execution. Hollifield et al. (2003) first estimate the expected time-to-execution, and then transform it to the probability of execution expected before the end of the next trading day.¹⁰ This representation loses much information relative to using the expected time-to-execution itself. Indeed, our results indicate that the probability of execution loses much of its explanatory power once the expected time-to-execution is included in the model.

3 The Model

To test the hypothesis that the expected time-to-execution affects the order submission strategy, we must deal with several econometric issues. First, Foucault et al. (2005) theoretical model makes it clear that if the demand

¹⁰We abstract from cancellations in their paper to illustrate the point. Adding cancellations into the picture does not change the logic.

for immediacy plays a role in order submission strategies, then one cannot separate the order submission strategy from its expected time-to-execution, since they are both endogenous variables. Second, the execution time is observed only for choices that were actually made, which requires us to deal with a selection bias in this variable. Finally, order cancellations force us to account explicitly for censored data. In this section we present an econometric model, which captures the intuition of the theoretical model, while addressing the econometric complications.

We build on a switching regressions model with endogenous switching described in Lee (1978) and Maddala (1983), and modify it to fit censored lifetime variables, as required for the analysis of our data. In this section we sketch the econometric model, while delegating the detailed derivation to the Appendix. The expected times-to-execution (or lifetime variables) and the order submission strategy (or price-aggressiveness) are endogenous variables in our model. The price-aggressiveness is represented by a discrete variable with three levels of aggressiveness - a market order, a price-improving limit order and a non-improving limit order. The expected time-to-execution of a market order is zero by definition, and we define only two lifetime variables, one for the price-improving and the other for the non-improving limit orders.¹¹

Consider a sequence of $i = 1, \dots, n$ orders, and denote by C_i a latent variable representing the equilibrium price-aggressiveness of order i . We assume that the price-aggressiveness depends on the expected times-to-execution and on a set of exogenous explanatory variables denoted by Z . We denote

¹¹While we use the term “time-to-execution”, a more accurate title for the endogenous lifetime variables is “time-to-first-fill,” which is the time till the first share of the order is executed. As we show below, the use of this variable allows us to assume that the order size has no direct effect on the time-to-execution. See also Lo et al. (2002).

by Y_1 the log time-to-execution of price-improving limit orders, and by Y_2 the log time-to-execution of limit orders that do not improve on the quoted price. We further assume that both lifetime variables depend on the same set of exogenous explanatory variables, denoted by X , which may overlap the set Z .

Let

$$Y_{1i} = \beta_1' X_i + v_{1i},$$

$$Y_{2i} = \beta_2' X_i + v_{2i},$$

and

$$C_i = \gamma' Z_i + \delta_1 E(Y_{1i}|X_i) + \delta_2 E(Y_{2i}|X_i) - u_i,$$

where

$$\begin{bmatrix} v_{1i} \\ v_{2i} \\ u_i \end{bmatrix} \sim N(0, \Sigma), \quad \Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{1u} \\ & \sigma_2^2 & \sigma_{2u} \\ & & \sigma_u^2 \end{bmatrix}.$$

The system of equations above is designed to capture the trade-off between the time-to-execution and the price aggressiveness. Each level of price aggressiveness results in a different time-to-execution, so there is one equation for each (non-zero) lifetime, Y_1 and Y_2 . The third equation is for the level of price-aggressiveness of the order, C , which depends exogenous variables and on the expected lifetimes of all possible order strategies.

Price priority rules in order driven markets force the execution time of an aggressive limit order to be shorter than that of a less aggressive one. This property should translate into the relationship $\hat{Y}_1 \leq \hat{Y}_2$ between the estimated lifetimes, for every order i . Furthermore, assuming that traders indeed value the execution time, Foucault et al. (2005) show that their price-aggressiveness increases in the expected times-to-execution. For example,

the probability of submitting a market order should be higher when the trader expects longer execution times for limit orders. This predicts positive signs of the coefficients of the expected lifetime variables, $\delta_1, \delta_2 > 0$.

Neither of the dependent variables defined above, Y_1 , Y_2 , and C , are directly observable. Moreover, since a significant percentage of the orders are cancelled before the execution, we are only able to observe censored lifetime values. Following Lo et al. (2002), we assume that all cancellations taking place before any fraction of the order is executed, represent a naive censoring of time-to-execution. Finally, since we use a discrete version of the price-aggressiveness variable,¹² the parameters γ , δ_1 , and δ_2 are estimable only up to a proportionality factor. We denote these variables by γ^* , δ_1^* , and δ_2^* respectively, and the scaled aggressiveness level is denoted by C^* .

Let us define the estimable form of our model:

$$I_i = \begin{cases} 0 & \text{if } -\infty < u_i^* < \gamma^* Z_i^* + (\delta_1^* \beta_1' + \delta_2^* \beta_2') X_i \\ 1 & \text{if } \gamma^* Z_i^* + (\delta_1^* \beta_1' + \delta_2^* \beta_2') X_i \leq u_i^* < \gamma^* Z_i^* + (\delta_1^* \beta_1' + \delta_2^* \beta_2') X_i + \mu_2^* \\ 2 & \text{if } \gamma^* Z_i^* + (\delta_1^* \beta_1' + \delta_2^* \beta_2') X_i + \mu_2^* \leq u_i^* < +\infty \end{cases} ,$$

$$S_i = \begin{cases} 1 & \text{if observation } i \text{ is censored} \\ 0 & \text{if observation } i \text{ is not censored} \end{cases} .$$

and

$$Y_i = \begin{cases} 0 & \text{if } I_i = 0 \\ \beta_1' X_i + v_{1i} & \text{if } I_i = 1 \text{ and } S_i = 0 \\ \beta_2' X_i + v_{2i} & \text{if } I_i = 2 \text{ and } S_i = 0 \end{cases} ,$$

¹²We use a discrete specification as do most studies on order submission strategy. One reason is the existence of the discrete tick size. A comprehensive discussion of the applicability of an ordered probit model in similar cases can be found in Hausman et al. (1992). This specification allows us to pull together observations with different bid-ask spreads or tick-size regimes. This makes it possible to detect the qualitative effect of price aggressiveness, which could have been swamped by noise in a small sub-sample. See further discussion in the Appendix.

where¹³

$$\begin{bmatrix} v_{1i} \\ v_{2i} \\ u_i^* \end{bmatrix} \sim N(0, \Sigma^*), \quad \Sigma^* = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{1u^*} \\ & \sigma_2^2 & \sigma_{2u^*} \\ & & 1 \end{bmatrix}.$$

We use a Full Information Maximum Likelihood (FIML) procedure to estimate our model.¹⁴ First, we use our results to learn whether the expected saving in time-to-execution increases the level of price aggressiveness, i.e. $\delta_1^*, \delta_2^* > 0$. Then we turn to the evaluation of the role of immediacy considerations under different regimes of information asymmetry. We consider this role to be significant if the results for stocks and government bonds, which are traded on the same platform on TASE, turn out to be qualitatively similar. Finally, we are interested in testing additional predictions derived from the theoretical models. We are interested in the variables that can potentially affect the expected time-to-execution (in X) and the order submission strategy (in Z). In particular, we are interested in the effects of competition between the suppliers of immediacy and of the arrival rate of traders on the expected time-to-execution. We are also interested in the effects of the spread size, the tick size and the expected times-to-execution on price-aggressiveness of submitted orders. A detailed discussion of each one of these variables and the empirical predictions derived from the theoretical models is presented below together with their definitions.

¹³Note that for $S_i = 1$, we only know that $Y_i > \beta_j' X_i + v_{ji}, j = 1, 2$.

¹⁴For a discussion of the FIML procedure and its relation to the “two-step” estimation procedures see Greene (2003, page 508).

4 Data, Sample and Variables

4.1 TASE Trading System

Tel Aviv Stock Exchange (TASE) at the time of our analyses was an electronic limit order book without intermediaries. Trading started with a call auction at 9:45 AM; a continuous phase between 9:45 AM and 4:45 PM, and a closing phase between 4:45 PM and 5:00 PM. The day starts with an empty order book at 8:30 AM, when traders start submitting either limit or market orders. At 9:45 AM all the submitted orders are crossed using an auction mechanism with time and price priority rules, and the continuous trading phase begins. The continuous phase is a non-intermediated limit order book. All traders observe the best three prices and quantities on each side. Traders may post either market or limit orders, and those are executed according to price and time priority rules. At 4:45 PM the closing phase begins; it is a simple crossing of traders' market orders that are executed, if possible, at the closing price. All unexecuted orders are cancelled at the end of the trading day, and the next day's book is empty till the first orders come in.

Some firms in the sample are cross-listed on European and US markets, however due to time difference, simultaneous trading of those stocks on TASE and on US venues takes place only during the last hour of the TASE trading session.

Equities, bonds, index options and futures contracts are all traded on the same TASE platform, with some minor differences in hours, minimum order size and tick size. The continuous trading phase for bonds starts at 10:45 AM and ends at 4:35 AM, which allows traders to devote exclusive attention to each instrument at the times of opening and closing. In this

study, we use data of the continuous trading phase for the samples of stocks and government bonds.

4.2 Data and Sample

We have chosen a sample period of three months, May 1, 2000 to July 31, 2000 for three reasons: the data for this period was relatively clean with the least missing records; there were no drastic events affecting the TASE market, and there were no significant regulatory changes.

We chose the stocks with the largest volume on TASE prior to the sample period, and kept those for which the tick size regime did not change.¹⁵ We further removed all the stocks, which did not have trading volume for at least 60 out of the 63 trading days in the sample and a median daily trading volume of at least 1,000,000 NIS. The remaining 32 stocks with some descriptive statistics are presented in **Table 1** (Panel A).

The sample of government bonds includes the most liquid bonds that were traded during the entire sample period,¹⁶ and belong to one of three categories: nominal bonds, inflation-linked, and dollar-denominated bonds. The list of 45 bonds with descriptive statistics are presented in **Table 1** (Panel B). All bonds have a tick size of 0.0001 NIS (New Israeli Shekels), which is finer than the minimal tick size category for stocks.

TASE had provided us with data on order submissions, cancellations, limit order book status, and trades for all stocks and government bonds for the year 2000. Each order in our dataset is time stamped and matched with the book status prior to its submission and with the subsequent trades

¹⁵The tick size depends on the price of the stock. TASE had four different tick sizes for stocks: 0.001, 0.01, 0.10, and 1 NIS.

¹⁶Each bond had to have a 3 months total trading volume of at least 10,000,000 NIS, a median daily trading volume of at least 50,000 NIS, and a median daily number of transactions of at least 5. We had also repeated the analyses with a stricter filter, using only bonds with at least 1.5 Million NIS median daily volume. The results were the same.

and/or cancellation. Note that even though we have data on all trades for each order, in this study we concentrate on the analysis of the duration till the first transaction or cancellation (“time-to-first-hit”, as in Lo et al. (2002)).

We use orders submitted during the continuous phase, half an hour after the opening trade and until the closing procedure at 16:45.¹⁷ Less than 3% (14,773) of observations in the sample of stocks and less than 4% (3,124) of observations in the sample of bonds are erroneous, thus excluded from our analyses.¹⁸ To maintain consistency of definitions for all order submission strategies, we exclude orders that were submitted when the spread was equal to one tick, since the price improving limit order is not a viable strategy in this case (140,520 observation are excluded from the sample of stocks and 8,045 from the sample of bonds).

In the continuous trading phase traders may submit two types of orders, either market or limit, but pure market orders are very rare - less than 0.5%. Instead, traders on TASE submit marketable limit orders, i.e. limit buy orders priced at or above the best ask or limit sell orders priced at or below the best bid. For the purposes of our analyses we treat these orders as if they were market orders, since the main intent is clearly to obtain immediacy. About one third of observations are categorized as market orders.

All unexecuted orders on the TASE are cancelled automatically at the end of the trading day yet, some of the orders are actively cancelled by the

¹⁷We define some of the explanatory variables as functions of the order flow or transaction rate half an hour prior to the submission of the order. Therefore, we only use orders submitted half an hour after the beginning of the continuous trading phase.

¹⁸Most errors are due to a missing trade records. Other errors are: market or marketable order that was not executed immediately; a record of the limit order book with the best bid exceeding the best ask. We also look for multiple orders submitted simultaneously (within the same 1/100 of a second), and remove all but the first one (less than 0.1% of the observations).

traders beforehand. On average, active cancellations occur much faster than the end-of-the-day cancellations, but the phenomena of fleeting limit orders observed by Hasbrouck and Saar (2004) for the Island ECN does not exist on the TASE.¹⁹ The distribution of time-to-cancellation by order type is presented in **Table 2**. On average, orders in the sample of stocks (Panel A) are cancelled faster than those in the sample of bonds (Panel B), and time-to-cancellation is longer for the less aggressive orders in both samples.

Table 3 presents the distribution of the “time-to-first-hit”, which is either an execution of the first share or a cancellation. 21% of the orders in the sample of stocks (Panel A) are price improving limit orders. As expected, they stay in the book for less than 1/3 of the time that the non-improving limit orders do. In the sample of bonds (Panel B), we observe price-improving limit orders more frequently (34%), yet on average, all limit orders in this sample spend more time in the book before they are filled or cancelled.

4.3 Definitions of Variables

Let p denote the limit order price in New Israeli Shekels (NIS), and q the order size as a number of shares. We also define p_{a1} and p_{b1} as the lowest ask and the highest bid at the time of the order arrival. Finally, q_{a1} and q_{b1} denote the depth in shares at the best ask and best bid.

Next, we define three **dependent variables**. The first is a discrete variable indicating the level of price aggressiveness. A market order is considered the most aggressive while non-improving limit order is considered the least

¹⁹Hasbrouck and Saar (2004) find that “fleeting” limit orders, that are cancelled within two seconds of submission, constitute almost 30% of the limit orders submitted to the Island ECN. This suggests a computerized trading strategy that demands rather than supplies immediacy using limit order strategy. The distribution of cancellation times on TASE suggests manual, rather than computerized strategies.

aggressive.

$$I = \begin{cases} 0 & \text{if market order,} \\ 1 & \text{if price-improving limit order,} \\ 2 & \text{if non-improving limit order.} \end{cases}$$

Obviously, there is more than one way to define the categories of this variable and we discuss this issue in detail in the Appendix. We would like to stress that our goal in the construction of this variable was to capture the main differences in price aggressiveness, without losing too many observations in the process and without losing the ability to draw meaningful conclusions from the analysis.

We denote by Y_j the log “time-to-first-hit” for a limit order of type $j = 1, 2$. We measure the lifetime of the order (in minutes) from the moment it is received on the exchange till the execution time of the first share. A cancellation of the limit order, before any fraction is executed, is considered to be naive censoring of the lifetime variable.²⁰

Next, we define the **explanatory variables** and discuss their predicted effects on the order-aggressiveness and the expected time-to-execution - these are summarized in Table 8 for easy reference. Our definitions refer to buy-side orders, but apply to sell-side orders with appropriate adjustments. Also note that all the discrete variables are appropriately transformed into indicator variables.

Measures of the arrival rate and the degree of competition

All the explanatory variables in this group are assumed to have a direct effect on the times-to-execution of limit orders, but only an indirect effect on the order submission strategy.²¹ We expect traders to submit aggressive

²⁰Lo et al. (2002) estimate both, time-to-first-fill and time-to-completion, and treat cancellations as naive censoring.

²¹For identification purposes we only need to exclude two explanatory variables from the price aggressiveness equation (see Maddala 1983, page 239) but we exclude eight. The

orders when the expected times-to-execution of the less aggressive order strategies are high. Thus, a positive effect on the time-to-execution should translate into an indirect positive effect on the price aggressiveness.

$SameLMT = \log(p_{a1} \cdot q_{a1})$ is the log of the monetary size of the depth at the best bid.

$OppLMT = \log(p_{b1} \cdot q_{b1})$ is the log of the monetary size of the depth at the best ask.

Our prediction for $SameLMT$ and $OppLMT$ are based on the results of Parlour (1998). If the buy-side depth is large, the expected time-to-execution of a non-improving buy-side limit order is high. On the other hand, when the opposite-side depth is high, the expected time-to-execution of non-aggressive orders on the opposite-side is long. That will encourage the sell-side traders to submit aggressive orders which will lead to shorter time-to-execution for subsequent buy-side limit orders. High values of the variable $SameLMT$ and low values of $OppLMT$ should both lead to a longer expected time-to-execution for non-aggressive orders.

$SameMKT$ is the ratio of the monetary value of buy-side market orders, submitted in the last half hour, to the total monetary value of buy-side orders in the same time interval.

$OppMKT$ is the ratio of the monetary value of sell-side market orders, submitted in the last half hour, to the total monetary value of sell-side orders in the same time interval.

Both variables, $SameMKT$ and $OppMKT$, are proxies for the proportion of impatient traders in the population at the time of order arrival.

identification assumptions and some robustness checks are discussed in detail at the end of this section.

According to Foucault et al. (2005), a high proportion of the impatient sell-side traders, who tend to post market rather than limit orders, should reduce the expected time-to-execution of all limit orders on the buy-side. On the other hand, high proportion of impatient buy-side traders *SameMKT* should have exactly the opposite effect. It should reduce the expected time-to-execution of sell-side limit orders, lead to the use of less aggressive order strategies by the patient sell-side traders and, eventually, result in longer expected time-to-execution of the buy-side limit orders. Thus, high values of the variable *SameMKT* and low values of *OppMKT* should both lead to a longer expected time-to-execution for all buy-side limit orders.

ArrivalRate is the log of the monetary value of orders, submitted in the last half hour. This variable is a proxy for the arrival rate of traders, and according to Foucault et al. (2005) should reduce the expected time-to-execution.

TradeVolume is log of the volume traded in the last half hour prior to order submission. Even in the presence of the order arrival rate variable, the trade volume variable is not redundant. It serves as a complementary measure, since not all the submitted orders are executed. Again, we expect *TradeVolume* to have a negative impact on the expected time-to-execution.

Imbalance is the difference between the monetary value of the same-side and the opposite-side orders, scaled by the total monetary size of orders, all measured in the last half hour.

$$LastTrade = \begin{cases} 1 & \text{if the previous trade was buyer initiated,} \\ 0 & \text{otherwise.} \end{cases}.$$

Both *LastTrade* and *Imbalance* are measures of the competition between traders on the same side of the market. *Imbalance* takes values between -1 and +1, which represent the magnitude of order imbalance. For

a buy-side order, *Imbalance* is positive when the total value of buy-side orders in the last half an hour is larger than that of the sell-side orders, and negative when the situation is reversed. The variable *LastTrade* is a more recent measure of the same property. There is no theoretical prediction for those variables, since most models use a symmetric setting for buyers and seller. Intuitively, just like the imbalance between demand and supply of immediacy, positive value of *Imbalance* and *LastTrade* should lead to a longer expected time-to-execution of buy-side limit orders.

Variables determining the trade-off between the expected time-to-execution and the price aggressiveness

$Spread = \log(p_{a1} - p_{b1})$ is the log of the inside quoted spread before the order is submitted.

$$Tick = \begin{cases} 0.001 & \text{if } p < 5, \\ 0.01 & \text{if } 5 \leq p < 50, \\ 0.1 & \text{if } 50 \leq p < 500, \\ 1 & \text{if } p \geq 500. \end{cases}$$

is the minimum price improvement in NIS.

$Lifetime_j = \beta'_j X$, $j = 1, 2$ is the expected log lifetime, in minutes, for price-improving or non-improving limit orders. Note that for market orders the expected lifetime is zero by definition.

The variables in this group capture the trade-off between the expected execution time and price, which defines the trader's order submission strategy in the Foucault et al. (2005) theoretical model.²² The spread, tick-size and expected times-to-execution are all cost components that the trader faces when submitting the order, and we expect the optimal order to minimize the sum of all those costs. Thus, we expect the trader to post a

²²The same trade-off is modeled by Rosu (2006) under the assumptions of a zero tick-size and costless cancellations.

price-aggressive order when the spread-size is small, the tick-size is low and the expected lifetimes of non-aggressive orders are long.

Control variables for intraday and day-of-the-week patterns

We know from Admati and Pfleiderer (1988) that time of the day may have a crucial effect on the trading choices; many empirical studies also document patterns of spread, volume, and volatility changes over the course of the day (e.g., Harris (1986), Jain and Joh (1988), McInish and Wood (1990) among many others). Therefore, we must control for this variation.

$$Daytime = \begin{cases} 10:15 - 10:45 & \text{if order is posted at 10:15 - 10:45,} \\ 10:45 - 11:45 & \text{if order is posted at 10:45 - 11:45,} \\ \vdots & \\ 15:45 - 16:45 & \text{if order is posted at 15:45 - 16:45.} \end{cases}$$

is a discrete variable that indicates the time of order submission during the continuous trading phase. The corresponding indicator variables control for the intraday effects.

$$Weekday = \begin{cases} LastDay & \text{if last day of the trading week,} \\ FirstDay & \text{if first day of the trading week,} \\ MidWeek & \text{otherwise.} \end{cases}$$

is a variable that controls for intraday effects. The degree of impatience may differ during different weekdays, thus we examine whether they have an impact on execution times and on order submission strategies.

IVolatility is the average price range in the last half hour prior to order submission, scaled by the range midpoint. The range is the difference between the maximum and the minimum midpoint prices in that time interval.²³

²³This measure of volatility overcomes the bias induced by the bid-ask bounce within different tick-size regimes (see Hau (2003)). The same applies to the intraday volatility below.

Cross sectional and daily control variables

The following variables are expected to capture the variation across stocks and the variation of market conditions across days, since we pool all the orders in all the stocks (see also Lo et al. (2002)).

Price is the log of the average opening price (NIS) of the stock; *Volume* is the log of the average daily volume (NIS) in the stock; and *Volatility* is the average of the daily price range, scaled by the range midpoint. The range itself is the difference between the maximum and minimum midpoint prices in the continuous trading phase.

ZVolume is the *Z* -score of the volume traded at the opening auction. We calculate the volume for each stock and day, and then subtract the sample-period mean and scale by the sample-period standard deviation for each stock.

We expect the time-to-execution of limit orders to be shorter when the volume and volatility are high, since in those cases any limit order is more likely to be hit shortly after submission. As for the effect of volatility on price aggressiveness, Foucault (1999) predicts that a high level of volatility should decrease price aggressiveness, since the picking off risk increases. On the other hand, Hasbrouck and Saar (2002) document that a high level of volatility has a positive effect on price aggressiveness and lead to an increase in the proportion of market orders in the order flow.

TA25 is an indicator variable for stocks in the index of the 25 largest stocks. *DualList* is an indicator variable for stocks also traded on a foreign exchange. Out of the 32 stocks in our sample, 18 are included in the TA25 index, 11 are traded on another exchange, and 6 stocks are included in both groups.

Other variables

Sell is the indicator variable for sell orders. In some empirical studies, sellers are documented to be more aggressive than buyers, yet in most theoretical studies buy and sell-side orders are modeled symmetrically.

$OrderSize = \log(pq)$ is the log of the monetary order size (NIS). We use the order size as a control variable, since most theoretical models deal with fixed size orders. The effect of this variable on price aggressiveness could go either way. A large order may drive the trader to ensure execution by using an aggressive strategy. On the other hand, such a trader might not want to expose his demand for immediacy and make a costly price concession, thus he may prefer a less aggressive strategy.

$OrderSize * DayEnd$ is an interaction between the variable *OrderSize* and an indicator of the last hour of trade ($Daytime = 15:45 - 16:45$). We believe that at the end of the trading day the submitted order size may be more restricted than at any other time, since they are expecting a break in trade. Moulton (2005) finds that at the end of a quarter the restricted order size has a positive effect on price aggressiveness, and we expect to find the same effect at the end of the day. Therefore, if the order is large and the trader is restricted (trading at the end of the day) we expect an increase in price aggressiveness.

Round is an indicator variable that takes the value of 1 when the limit order price is a round multiple of five ticks. Harris (1991) shows that round numbers are observed more frequently than others, and since traders may tend to choose round limit order prices we need to control for that.

Special definitions for the sample of bonds

Most variables, their definitions and their predicted effects are the same for the stocks and the government bonds, yet, there are a couple of differences. All government bonds in our sample are traded exclusively on the TASE and are not included in any stock index. Therefore, the indicator variables *TA25* and *DualList* are not relevant for the bonds sample. The variable *Tick* is also omitted, since all bonds have the same tick size. On the other hand, we include indicator variables that separate nominal, CPI-linked, and dollar denominated bonds.

One more difference between the two samples arises from the fact that bonds start trading later than stocks, thus there are fewer indicator variables for the time of the day effects.

Identification

The model parameters are identified, since the set of explanatory variables in the lifetime equations is not the same as that in the price-aggressiveness equation. In our setting, we have to exclude at least two variables from the lifetime equations for identification purposes yet, based on theoretical models, we exclude more: *OrderSize*, *OrderSize*DayEnd*, *Round*, *Spread* and *Tick* from the lifetime equations, and *SameLMT*, *OppLMT*, *SameMKT*, *OppMKT*, *ArrivalRate*, *Imbalance*, *LastTrade* and *TradeVolume* from the price-aggressiveness equation. In **Appendix 2** we present detailed theoretical arguments for our exclusion restrictions, and the results of alternative specifications. We show that our main result that time-to-execution is an important determinants of the order-aggressiveness is intact under most specifications. However, under specifications that are not derived from theory, various other variables appear with a counterintuitive sign.

5 Results

First, we present the results of the Full Information Maximum Likelihood (FIML) estimation procedure for the two samples - stocks and bonds. Next, we use our results to calculate an estimate of the monetary value of time. Then, we show that the explanatory power of expected time-to-execution dwarfs that of the probability of execution by the end of the trading day. Finally, we show that the results are robust by considering a sample of stocks that change the tick size.

5.1 Stocks

The estimation results for the lifetimes of improving and non-improving orders are presented in **Table 4** (Panel A). As predicted in Parlour (1998) and Foucault et al. (2005), the depth of the book at the best bid (ask) - *SameLMT*, the proportion of market orders on the buy-side (sell-side) - *SameMKT*, and the order imbalance (*Imbalance* and *LastTrade*), all have positive and significant effects on the lifetimes of buy-side (sell-side) orders. At the same time, again as predicted, the depth of the book at the best ask (bid) - *OppLMT*, the proportion of market orders on the sell-side (buy-side) - *OppMKT*, the arrival rate of traders to the market (*ArrivalRate*) and the volume of executed orders (*TradeVolume*) have a negative and significant effect on the lifetimes of buy-side (sell-side) orders.

While the effects of the time and day control variables (*Daytime* and *Weekday*) are not always significant, they do exhibit interesting patterns. The expected execution time of non-improving limit orders decreases monotonically during the day. The expected execution time of price-improving limit orders is increasing over the course of the day. We also observe shorter lifetimes for all types of limit orders in midweek.

High trading volume and high volatility defined as cross-sectional (*Volume*, *Volatility*), daily (*ZVolume*), or intraday measures (*TradeVolume*, *IVolatility*) reduce the expected execution time of limit orders. These results are consistent with predictions as well.

Notice that the correlation coefficients between the lifetimes and the order aggressiveness (ρ_{ju} , $j = 1, 2$) are negative and significant for both types of limit orders. This result supports our hypothesis that there is selection bias in the choice of order submission strategy, and that the simultaneous specification of the lifetimes and the price aggressiveness choice is required. Furthermore, the negative sign of the coefficient means that price aggressiveness is positively correlated with the expected lifetimes of non-aggressive limit orders. In other words, aggressive orders are more frequent when the expected lifetime of non-aggressive orders is relatively longer. This finding is consistent with the models in Foucault et al. (2005) and Rosu (2006), where traders choose to submit price aggressive orders when the expected lifetimes of less aggressive orders are longer.

Our econometric approach makes it possible to estimate the expected lifetimes conditional on any one of the three possible order strategies.²⁴ Comparing the lifetime estimates, we find that for every order in our sample, the estimated expected lifetime of an aggressive order strategy is shorter than that of the less aggressive one. This result is consistent with the price priority rule of the exchange, even though we did not impose it as a restriction in the estimation procedure.

The estimation results for price aggressiveness are presented in **Table 5** (Panel A). The variables representing the trade-off between the expected execution times (*Lifetime_j*, $j = 1, 2$) and price (*Spread*, *Tick*) have sig-

²⁴ Actually, the time-to-first-hit of a market order is zero by definition.

nificant effects on the degree of price aggressiveness. Both variables that make a market order costly: a large spread and a large tick-size, reduce the probability of submitting a market order. On the other hand, long expected execution times for the less aggressive order strategies increase the probability of submitting a market order. This result supports the existence of a non-trivial effect of the expected execution time on price aggressiveness, corroborating the predictions of Foucault et al. (2005) and Rosu (2006).²⁵

The variables that control for order submission time (*Daytime* and *Weekday*) affect the order submission strategies in the predicted way. The probability of an aggressive order strategy increases towards the end of the trading day, as well as towards the end of the trading week. These results are consistent with the conjecture that traders become less patient before the end of continuous trading period. This result is consistent with the findings of Bosetti, Kandel and Rindi (2007), who study the effects of a Call Auction on the market quality during the continuous trading phase in Paris Euronext and in Milan.

The *OrderSize* has a negative effect on price aggressiveness; this could be due to the fact that the total monetary cost of the price concession are larger for a large order. As the rest of the literature, we do not observe order splitting, which may account for this effect as well. It could be that splitting is a substitute for price aggressiveness.²⁶ Following Moulton (2005)

²⁵In this sample the effect of the tick size is shown using cross-sectional comparisons, as by construction it varies only across stocks. We have repeated the analyses using a sample of stocks that changed the tick size over the course of our sample period. The effect of tick size on price aggressiveness remains strong in that setting as well. We also show that traders need some time getting used to the new tick size regime, as they submit somewhat less aggressive orders, when the tick size declines, and somewhat more aggressive when it increases. These results are available upon request.

²⁶Bessembinder et al. (2008) study the choice of the proportion of a hidden order to be displayed (a special case of the order splitting), in the context of Euronext-Paris. TASE, unlike Euronext, allows no hidden orders, thus we cannot identify multiple orders

we conjecture that order splitting is less prevalent towards the end of the day, when the submitted order size should be closer to the true desired order. We show that the interaction between the *OrderSize* and the last hour dummy has a positive effect on price aggressiveness, similar to findings in Moulton (2005). If a trader arrives at the end of the trading day with a relatively large order, his price aggressiveness is expected to increase to ensure execution.

Higher volume and volatility increase the probability of an aggressive order. The volatility effect is consistent with the prediction of Harris (1998) and the results of Hasbrouck and Saar (2002). In fact, our econometric approach makes it possible to separate two opposite effects of volatility discussed in Hasbrouck and Saar (2002). The mechanistic effect of high volatility results in low expected times-to-execution for limit orders and thus has an indirect negative effect on price aggressiveness. On the other hand, the direct positive effect of volatility on price aggressiveness can be attributed to an increase in the picking off risk.

As a side issue, we show that traders do tend to prefer round numbers, which corroborates the findings in Harris (1991).

5.2 Bonds

Estimation results for the lifetimes in the sample of bonds is presented in **Table 4** (Panel B). Most measures of depth, competition and activity affect bonds in the same way they affect stocks, as predicted by the immediacy-centered models. The coefficient of *Imbalance* is an exception and, counter-intuitively, implies that if more same-side orders arrive to the market then traders expect a shorter time-to-execution. For the sub-sample submitted by the same trader.

of non-improving limit orders the effect of this variable is not significantly different from zero, but for the price-improving sub-sample it is. We are not sure how to interpret this result.

The correlation coefficients (ρ_{ju} , $j = 1, 2$) are again negative and significant. We conclude that the simultaneous specification is required for bonds as well as stocks, since there exists a positive correlation between the expected time-to-execution and the level of price-aggressiveness. Furthermore, for every order in the sample of bonds, the expected lifetime of an aggressive order strategy is again shorter than that of the less aggressive one. Again, this result is consistent with the price priority rule of the exchange, even though we did not impose it as a restriction.

The estimation results for price aggressiveness for the sample of bonds are presented in **Table 5** (Panel B). While most results for the sample of bonds are qualitatively very similar to those obtained for the sample of stocks, we must caution against making quantitative comparisons of the marginal effects across the two samples, as there are scale effects.²⁷ In the sample of bonds, two of the variables representing the trade-off between the expected time-to-execution ($Lifetime_1$) and the price aggressiveness ($Spread$) have significant effects on the probability of submitting an aggressive order. A large spread, which makes the price-aggressive market order strategy costly, reduces the probability of choosing that strategy, while longer expected time-to-execution of a price-improving limit order increases that probability. The effect of the expected time-to-execution of a non-improving limit order ($Lifetime_2$) on the probability of submitting a

²⁷It is important to stress that those effects are estimated at the sample mean and depend on those values of the explanatory variables, which assume very different magnitudes across the two samples. The average bond order, for example, is about ten times bigger than the average stock order. Thus, an attempt to compare the marginal effects across samples is equivalent to the comparison of variables measured on different scales.

market order is negative, contrary to our expectation. We guess that the use of the same set of explanatory variables to estimate both expected times-to-execution ($Lifetime_1$ and $Lifetime_2$), and the fact that those variables affect the times-to-execution in a similar way, are the causes of the sign reversal. It is possible that the information content of the much smaller sample of bonds does not allow us to properly identify all the effects.²⁸

In summary, the results for the two samples provide answers to our three main questions. First, parameters determining the trade-off between the expected time-to-execution and the order aggressiveness have a significant effects on the probability of submitting aggressive orders. Indeed, longer execution times of non-aggressive orders increase the probability of using an aggressive order, while higher spread or tick-size reduce that probability. Second, the results corroborate the empirical predictions of theoretical models of the limit order book. The expected time-to-execution is long when the competition between suppliers of immediacy is intense, the arrival rate of traders is low, the depth of the book on the same side as the order is high, and the volume and volatility are low. Finally, our results are qualitatively similar for the samples of stocks and government bonds. This finding supports our hypothesis that immediacy-based considerations have a significant effect on the high frequency price dynamics in order driven markets, regardless of the degree of information asymmetry associated with the traded instrument.

²⁸Note that we start with a total sample size of 59,000 observations for the bonds. Less than 343,265 observations that we have for the stocks, but the sample size does not seem small. However, only the non-censored observations in the sub-sample of non-improving limit orders supply the information for the non-improving limit order lifetime estimation, and there we have only 4,352 observations.

5.3 How Much Money is Time Worth?

The results above are highly significant, yet do not indicate whether the effect of the expected time-to-execution on order submission strategy is economically significant. We demonstrate the economic magnitude of this effect for one liquid stock and one liquid bond. We choose TEVA, which is the highest volume stock on TASE and is also traded on NASDAQ, and a short-term government bond (equivalent to a T-Bill), which is also very liquid. We concentrate on very liquid securities to be conservative, as it biases the calculation against finding strong effect.

We use our parameter estimates to calculate expected times-to-execution and probabilities of choosing each one of the three order submission strategies at the sample mean.²⁹ Next, we assign a monetary cost to each order submission strategy, representing price concession. We consider the non-improving limit order as the baseline strategy and assign it the cost of zero, as it is set at the best possible price out of all possible orders. Relative to the baseline, the cost of a price improving limit order is at least one tick and the cost of a market order is at least the size of the spread - we use these conservative estimates as the costs of submitting these orders.

Now, suppose a representative trader arrives at the market. We assume that his order submission strategy is determined by the probability distribution estimated above. Therefore, the expected cost of immediacy is an average of the costs of the possible orders weighted by their respective probabilities. We normalize this cost by the average bid-ask spread in

²⁹Note that we use the characteristics of an average order, but we do not simply calculate sample average lifetimes and probabilities. For the characteristics of an average order, we use the estimates of model parameters to calculate the expected times to execution of every possible strategy and the probabilities of choosing each one of the three strategies. Our calculation produces maximum likelihood estimates since the maximum likelihood estimator of $f(\theta)$ is $f(\hat{\theta}_{ML})$.

this security. We repeat this calculation for an extreme case that assumes zero expected time-to-execution for any limit order. This can be seen as a robustness check of the model, since we expect to observe a very low cost attributed to execution delay in this case, even though the econometric model has no such restriction. In other words, if our model is correct, then zero delay should cause all traders to submit non-improving limits.

The results of our calculation are presented in **Table 6**. When the time-to-execution is set to zero, our results indicate that the expected price concessions are practically zero: 0.1 and 0.03% of the spread for the stock and bond respectively. On the other hand, evaluated at the sample mean, these costs are quite significant: for TEVA (Panel A) these costs are over 55% of the average bid-ask spread, and for the government bond (Panel B) they are 33% of the relevant spread.³⁰ This calculation demonstrates that the demand for immediacy accounts for a large proportion of the bid-ask spread, complementing the other, more traditional spread determinants.

5.4 Time vs. Probability of Execution

The extant literature implicitly assumes that traders care only about the probability of execution, whereas this paper shows that they care about the expected time to execution. The formulation in Hollifield et al. (2003) clearly shows that the expected probability-of-execution by the end of the trading day is a (non-linear) transformation of the expected time-to-execution. Our earlier discussion suggests that this transformation may lose much information.³¹ However, the question whether the additional information con-

³⁰We also show that the cost is a monotonically increasing and concave function of the expected times-to-execution. The results are not presented for brevity.

³¹An extended discussion of a similar issue and strong arguments in favor of estimating a lifetime / duration to event model rather than a probability of event model can be found in Heckman and Singer (1984).

tained in the time-to-execution measure is an important determinant of order aggressiveness is an empirical one. To test this, we estimate several variations of our model including both measures. We compare the estimation results in **Table 7**.

We define the probability of execution by the end of the trading day similarly to Hollifield et al. (2003). This involves estimating the expected time-to-execution and calculating the probability of execution before the end of the trading day.³² Since we have two expected lifetime estimates (one for price-improving and one for non-improving limit orders), we obtain two estimated execution probabilities.³³ The parameter estimates are presented in **Table 7**.

First, we estimate the order submission decision separately for lifetimes (model 1) and execution probabilities (model 2). Not surprisingly, the lifetime measures exhibit the expected effects: longer lifetimes of limit orders increase the likelihood of a price aggressive order. Yet, the case of model 2 is different. One of the probability coefficients has a counter-intuitive sign,

Note that a model of the expected lifetime-to-execution can be easily transformed into a model of the execution hazard function - the probability of execution by any point in time. For details see Cox and Oakes (1984).

³²Hollifield et al. (2003) estimate an independent competing risks model for the expected time-to-execution and time-to-cancellation. Then, they calculate the probability of execution using parameter estimates of two distributions: lifetime to execution and lifetime to cancellation. We make a similar independence assumption, but do not explicitly model and estimate the expected time-to-cancellation. Therefore, we simplify the probability calculation and use the trading rule of the exchange to determine the expected time-to-cancellation (on TASE all unexecuted orders are cancelled by the end of the trading day).

³³Since the expected probabilities are non-linear functions of the endogenous lifetime variables, the estimation procedure is more complicated. Simply using non-linear functions as explanatory variables in the aggressiveness equation would lead to "forbidden regression". We refer to Wooldridge (2001), sections 9.5 and 9.6, for a detailed discussion of this issue and the suggested estimation procedure that we follow.

In our model, given the increased dimensionality of the problem, we use a two-stage estimation procedure. It is worth mentioning that estimating our original model using both, the FIML and the two-stage procedures, obtains very similar results.

predicting an increase in price aggressiveness as the probability of execution increases. The spread coefficient in this case also changes the sign.

Next, we run a “horse race” between the two measures by including both in a nesting model (model 3): both sets of coefficients are significant, but one of the probability measures has a counterintuitive coefficient as in model 2. In addition, the likelihood ratio tests indicate that the probability of execution adds less information than the lifetimes.³⁴ We suspect that most of the explanatory power of the probability measure is due to its non-linear nature in time, especially towards the end of the day. To explore this hypothesis, we re-estimate the models above excluding the last trading hour of the day. The probability of execution has lost almost all its explanatory power, whereas the lifetime remain as significant as before.³⁵

In summary, our analyses in this section show that the expected time-to-execution carries more information relevant to explaining the order aggressiveness than the probability of execution by the end of the trading day. Traders seem to value not only one execution probability, but many of them, and this is the main reason for the informational advantage of the lifetime measure.

6 Conclusions

In this paper, we ask whether the expected time-to-execution for order strategies of varying levels of price-aggressiveness affects traders’ choices among these strategies. Based on recent theoretical models, we introduce an econometric framework for modeling order submission strategies and time-

³⁴We also ran a J-test comparing model 1 to 2 and visa versa. This test is a different approach to nesting both models in one general model and, not surprisingly, we get similar results.

³⁵Results are available from the authors upon request.

to-execution as a set of simultaneous equations, where both elements are endogenous and affect one another. We use a very detailed data set from the Tel Aviv Stock Exchange that offers a perfect setting to test such models. Our empirical results confirm the existence of the expected trade-off between the expected time-to-execution and the level of price-aggressiveness, and offer a characterization of the relationship between those variables and other exogenous explanatory variables that affect the trade-off. We show that the expected time-to-execution is a more informative measure for the investor's impatience than the probability of execution, which is typically used in the extant literature. According to our calculations, the monetary cost associated with the expected time-to-execution in our sample can account for over one half of the bid-ask spread. Finally, we obtain qualitatively similar results for stocks and government bonds, which are traded on the same platforms on TASE. This result strengthens our claim that immediacy-based considerations have a significant effect on high-frequency price dynamics regardless of the information asymmetry regime. Overall, our results suggest that the demand for immediacy is an important channel for price determination, and should play a more prominent role in the literature.

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Appendix 1: The Econometric Model

Consider a sequence of $i = 1, \dots, n$ order decisions, and denote by C_i a latent variable representing the price-aggressiveness of order i . We assume that the price-aggressiveness depends on the expected times-to-execution and on a set of exogenous explanatory variables denoted by Z . We denote by Y_1 the log time-to-execution of price-improving limit orders, and by Y_2 the log time-to-execution of limit orders that do not improve on the quoted price. We further assume that both lifetime variables depend on the same set of exogenous explanatory variables, denoted by X , which may overlap the set Z .³⁶

Let

$$Y_{1i} = \beta_1' X_i + v_{1i},$$

$$Y_{2i} = \beta_2' X_i + v_{2i},$$

and

$$C_i = \gamma' Z_i + \delta_1 E(Y_{1i}|X_i) + \delta_2 E(Y_{2i}|X_i) - u_i,$$

where

$$\begin{bmatrix} v_{1i} \\ v_{2i} \\ u_i \end{bmatrix} \sim N(0, \Sigma), \quad \Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{1u} \\ & \sigma_2^2 & \sigma_{2u} \\ & & \sigma_u^2 \end{bmatrix}.$$

The system of equations above is designed to capture the order decision process, yet all the dependent variables, Y_1 , Y_2 and C , are not observable. Instead, we are only able to observe the price aggressiveness as a discrete variable that we denote by I ,

³⁶Note that the set X may overlap the set Z but not completely. There must be two variables in X that are not included in Z for identification reasons.

$$I_i = \begin{cases} 0 & \text{if } 0 < C_i < +\infty \text{ (market order - most aggressive)} \\ 1 & \text{if } -\mu_2 < C_i \leq 0 \text{ (price-improving limit order)} \\ 2 & \text{if } -\infty < C_i \leq -\mu_2 \text{ (non-improving limit order)} \end{cases},$$

or, defined in terms of the distribution of the residual u_i ,

$$I_i = \begin{cases} 0 & \text{if } -\infty < u_i < \gamma'Z_i + \delta_1 E(Y_{1i}|X_i) + \delta_2 E(Y_{2i}|X_i), \\ & \text{if } u_i \geq \gamma'Z_i + \delta_1 E(Y_{1i}|X_i) + \delta_2 E(Y_{2i}|X_i) \text{ and} \\ 1 & \quad u_i < \gamma'Z_i + \delta_1 E(Y_{1i}|X_i) + \delta_2 E(Y_{2i}|X_i) + \mu_2, \\ 2 & \text{if } u_i \geq \gamma'Z_i + \delta_1 E(Y_{1i}|X_i) + \delta_2 E(Y_{2i}|X_i) + \mu_2. \end{cases}.$$

The lifetime variables are observed only for the order strategy that is eventually chosen by the trader and, since a significant percentage of the orders are cancelled before they are executed, we observe a mixture of censored and non-censored values. Thus,

$$Y_i = \begin{cases} 0 & \text{if } I_i = 0 \\ Y_{1i} & \text{if } I_i = 1 \text{ and } S_i = 0 \\ Y_{2i} & \text{if } I_i = 2 \text{ and } S_i = 0 \end{cases},$$

where

$$S_i = \begin{cases} 1 & \text{if observation } i \text{ is censored} \\ 0 & \text{if observation } i \text{ is not censored} \end{cases}.$$

Finally, taking into account the discrete nature of the observable price-aggressiveness, we have to scale the variance of the error term, u_i , to 1. To see the effect of this assumption on other parameters we substitute the expected lifetimes into the equation of the price aggressiveness. If we present this equation in its reduced and scaled form, it is easy to see that the parameters γ , δ_1 , and δ_2 are estimable only up to a proportionality factor:

$$C_i^* = \gamma^* Z_i + (\delta_1^* \beta_1 + \delta_2^* \beta_2)' X_i - u_i^*,$$

where

$$C_i^* = \frac{C_i}{\sigma_u}, \gamma^* = \frac{\gamma}{\sigma_u}, \delta_j^* = \frac{\delta_j}{\sigma_u} \ (j = 1, 2) \text{ and } u_i^* = \frac{u_i}{\sigma_u}.$$

The Full Information Maximum Likelihood Procedure

Following the study of Lo et al. (2002) we assume that all cancellations that occur before any fraction of the order is executed, represent naive censoring of the lifetime values. That is, if $S_i = 1$ and the value of Y_i is censored we can only say that $Y_i > \beta'_j X_i + v_{ji}$, $j = 1, 2$. Under this assumption, the likelihood function of the model is as follows:

$$\begin{aligned} L(\gamma^*, \mu_2^*, \delta_1^*, \delta_2^*, \beta_1, \beta_2, \sigma_1^2, \sigma_2^2, \sigma_{1u^*}, \sigma_{2u^*}) = & \\ & \prod_{\substack{I_i=0 \\ S_i=0}} \int_{-\infty}^{\gamma^{*'} z_i + (\delta_1^* \beta_1 + \delta_2^* \beta_2)' x_i} \phi(u_i^*) du_i^* \\ & \prod_{\substack{I_i=1 \\ S_i=0}} \int_{\gamma^{*'} z_i + (\delta_1^* \beta_1 + \delta_2^* \beta_2)' x_i}^{\gamma^{*'} z_i + (\delta_1^* \beta_1 + \delta_2^* \beta_2)' x_i + \mu_2^*} f_1(y_i - \beta_1' x_i, u_i^*) du_i^* \\ & \prod_{\substack{I_i=2 \\ S_i=0}} \int_{\gamma^{*'} z_i + (\delta_1^* \beta_1 + \delta_2^* \beta_2)' x_i + \mu_2^*}^{+\infty} f_2(y_i - \beta_2' x_i, u_i^*) du_i^* \\ & \prod_{\substack{I_i=1 \\ S_i=1}} \int_{(y_i - \beta_1' x_i)}^{+\infty} \int_{\gamma^{*'} z_i + (\delta_1^* \beta_1 + \delta_2^* \beta_2)' x_i}^{\gamma^{*'} z_i + (\delta_1^* \beta_1 + \delta_2^* \beta_2)' x_i + \mu_2^*} f_1(v_{1i}, u_i^*) du_i^* dv_{1i} \\ & \prod_{\substack{I_i=2 \\ S_i=1}} \int_{(y_i - \beta_2' x_i)}^{+\infty} \int_{\gamma^{*'} z_i + (\delta_1^* \beta_1 + \delta_2^* \beta_2)' x_i + \mu_2^*}^{+\infty} f_2(v_{2i}, u_i^*) du_i^* dv_{2i} \end{aligned}$$

where $f_j(\cdot, \cdot)$ are the bivariate normal *pdfs* of (v_{ji}, u_i^*) , $j = 1, 2$.³⁷

³⁷In an earlier version of this paper we used a three stage estimation procedure rather than the FIML to estimate the model parameters. The results were practically similar to those achieved using the FIML. The advantage of the three stage procedure is the rather easy and fast convergence of the optimization algorithm and the higher certainty of finding the global maximum. The problem with that estimation procedure was that the assumption of normality of the censored log lifetime variables was not accurate. This problem undermined the consistency of the estimates in the lifetime equations, which is a crucial assumption in a multistage procedure.

Categories of the Order Submission Strategy Variable

The discrete variable I , which represents the order submission strategy in our model, has three categories: market, price improving limit and non-improving limit order. Considering our use of an ordered and discrete variable, one could argue that it is not complicated to slice the latent variable C into more categories and make finer distinctions between order strategies. Nevertheless, the slicing cannot be too fine, since it also determines the size of each sub-sample used to estimate the expected lifetime conditional on the choice of order strategy. Yet, another reason for the limited number of categories is that the finer slicing is also costly in terms of the total sample size. Note that in the three categories framework, all the strategies are feasible only if the spread size is at least two ticks. Thus, in the sample of stocks, 140,520 (26%) observations were excluded from our analyses and in the sample of bonds 8,045 (9%) were excluded for that reason. Had we used a four categories definition, adding the distinction between orders that improve by one tick and orders that improve by more than that, we could only use observations with an inside spread of three ticks and up. This detailed definition would have lead to an additional loss of 73,927 observations in the sample of stocks and 6,542 in the sample of bonds. Thus, we have chosen the three-categories definition for the discrete order strategy variable.

Appendix 2: Identification

In our setting, we have to exclude at least two variables from the lifetime equations for identification purposes yet, for theoretical reasons, we decided to exclude more. We exclude the variables *OrderSize*, *Round*, *Spread* and *Tick* from the lifetime equations, and *SameLMT*, *OppLMT*, *SameMKT*, *OppMKT*, *ArrivalRate*, *Imbalance*, *LastTrade* and *TradeVolume* from the price-aggressiveness equation. Below we present theoretical arguments for our exclusion restrictions, and the results of robustness tests of our specification.

First, we discuss the exclusion restrictions imposed on the lifetime equations. While *OrderSize* clearly determines the execution time of the entire order (time-to-completion), we believe that it doesn't have a direct effect on the execution time of the first share of that order (time-to-first-hit). Thus, we include a direct effect of *OrderSize* only in the price aggressiveness equation, through which it has an indirect effect on the lifetime. Foucault et al. (2005) and Rosu (2006) show that the size of the bid ask spread and the tick size should affect the lifetimes only through the price aggressiveness decision. In Foucault et al. (2005) the spread and the tick size are two key direct determinants of the optimal order submission strategy (Propositions 1-3), but does not have a direct effect on the expected lifetimes (Proposition 4). The only effect of on the lifetimes is an indirect one through the trade-off between the cost of price improvement and cost of waiting (page 6 equation 1). We are not aware of a theoretical argument supporting a direct effect of *Spread* or *Tick* on the lifetimes. We make the same assumption of indirect effect for the indicator variable *Round*, which may capture the behavioral bias of a trader submitting an order. Again, we are not aware of a reason

to include this indicator in the lifetime equations.

Our assumptions concerning the variables *OrderSize* and *Spread* may be controversial. We would like to point out that the identification of our model parameters does not rely on this exclusion restriction and therefore, we can reestimate the model with *OrderSize* and *Spread* in the lifetime equations. We present the results of this estimation of the lifetime equations in **Table A1**: in the first specification we include *OrderSize*, but exclude *Spread*: our original results hold for all the other explanatory variables, but the *OrderSize* reduces the lifetime of an order, which is counter-intuitive. This is because if the time-to-first-hit is positively correlated with the time-to-completion of the same order, larger orders should take longer to execute. In the second specification, both *OrderSize* and *Spread* are included in the lifetime equations and, as a result, many of the estimates change their signs, and we are at loss on how to explain these counterintuitive effects. For example, variables capturing higher arrival rates and more competing activity increase time-to-execution, instead of reducing it. Since our decision to exclude *OrderSize* and *Spread* from the lifetime equations is based on theoretical arguments and yields much more plausible results, we feel that this is the correct approach.

Next we discuss the exclusion restrictions imposed on the price aggressiveness equation. Parlour (1998), Foucault et al. (2005), and Rosu (2006) show that all variables that measure the characteristics of the order flow and the competition among traders have a direct effect on the expected execution time, but only an indirect one on price aggressiveness. In the context of Foucault et al. (2005) we proxy for the proportion of impatient traders in the population (θ_I) by the proportions of market orders (*SameMKT*

and *OppMKT*) in the last half hour before the order submission. Foucault et al. (2005) show that θ_I directly affects the expected lifetime (Proposition 4), but has only an indirect effect on price aggressiveness (Equation 1). The same holds for the arrival rate of traders (λ), which we proxy by the *ArrivalRate*. Another example is Parlour (1998), where the trader’s choice is affected by the depth of the limit order book. We measure for the bid and ask depth (b^B and b^A) by the observed order volume at the bid and ask sides (*SameLMT* and *OppLMT*). In Parlour (1998), the depth variables have a direct effect on the execution probabilities (Proposition 1), but only an indirect effect on price aggressiveness. The conclusions from the theoretical models of Rosu (2006) and Foucault (1999) are similar.

Finally, we deviate from the theoretical setting and reestimate our original model excluding only the required two (rather than eight) competition variables from the price aggressiveness equation. The results are presented in **Table A2**. Our original estimates are presented as **Model 1** and the new estimates as **Model 2**. Practically all the added explanatory variables show counter-intuitive effects on price aggressiveness. One example is the negative effect of the buy-side depth (*SameLMT*) and the positive effect of the sell-side depth (*OppLMT*) on price aggressiveness of buy-side orders, contrary to the prediction in Parlour (1998) and ample empirical evidence. Another example is the positive effects of *ArrivalRate*, and *TradeVolume* contrary to the prediction in Foucault et al. (2005). Clearly, once the effect of those variables on the expected time-to-execution is taken into account (included in the explanatory variables *Lifetime*₁ and *Lifetime*₂), there is no room for an additional direct effect on price aggressiveness.

Table 1, Panel A - Stocks

Summary statistics for the sample of 32 most liquid stocks traded on TASE, with no changes in the tick-size, for the sample period May 1, 2000 to July 31, 2000. The value of 'TA25 Indicator' is 1 if the stock is included in the TA25 index and the value of 'Dual Listing' is 1 if the stock is traded on some other venue in the US or Europe. We report the median daily NIS volume, the tick-size in NIS, the average bid ask spread (percentage spread) and the average price (NIS) of the stock at the opening.

#	Ticker	TA25 indicator	Dual Listing	Median Daily Volume	Tick Size (NIS)	Average Spread (%)	Average Price (NIS)
1	ISRA	0	0	7,890,501	0.001	0.68	0.16
2	AVNR	0	0	3,432,130	0.001	0.58	0.33
3	MGDL	1	0	1,829,317	0.001	0.77	4.03
4	DEDR	0	0	1,687,587	0.001	0.76	2.23
5	POLI	1	0	23,892,66	0.01	0.17	11.95
6	LUMI	1	0	18,765,99	0.01	0.19	8.72
7	BEZQ	1	0	17,611,43	0.01	0.20	23.06
8	MAIN	1	0	4,208,252	0.01	0.42	8.86
9	MZRH	1	0	3,765,695	0.01	0.40	12.22
10	MSHV	0	0	2,732,504	0.01	0.68	29.12
11	ELBT	0	1	2,492,884	0.01	0.57	39.26
12	SAE	1	1	2,364,396	0.01	0.47	14.56
13	AGIS	0	0	1,267,826	0.01	0.78	36.34
14	TEVA	1	1	24,171,90	0.1	0.14	215.66
15	KOR	1	1	12,540,95	0.1	0.27	413.68
16	DISI	1	0	11,645,29	0.1	0.30	224.37
17	NICE	1	1	10,950,82	0.1	0.24	291.05
18	FORT	0	1	7,944,436	0.1	0.29	219
19	IDBD	1	0	7,828,695	0.1	0.45	177.67
20	ELRN	1	1	4,825,355	0.1	0.45	142.46
21	POIN	0	1	4,376,808	0.1	0.59	93.89
22	IDBH	1	0	4,104,989	0.1	0.59	152.26
23	LPMA	0	0	3,720,620	0.1	0.65	142.68
24	MATV	1	1	2,395,000	0.1	0.76	82.48
25	CLIS	1	0	1,900,726	0.1	1.04	64.31
26	ELEI5	0	0	1,565,533	0.1	0.81	213.56
27	ESLT	0	1	1,420,616	0.1	0.62	58.57
28	BRAN	0	0	1,256,225	0.1	0.90	118.51
29	DELT	0	1	1,238,641	0.1	0.81	87.28
30	BLSQ	1	0	1,016,786	0.1	0.99	56.76
31	CLEI	1	0	3,169,510	1	0.59	919.18
32	ILCO1	0	0	2,120,941	1	0.82	767.63

Table 1, Panel B - Bonds

Summary statistics for the sample of 45 most liquid government bonds traded on TASE, for the sample period May 1, 2000 to July 31, 2000. The value of 'Bond Type' is 'Nominal' if the payments are nominal, 'Index' if they are indexed and 'Dollar' if they are linked to the US Dollar. We report the median daily NIS volume and the average bid ask spread (percentage spread). All bonds in the sample have a tick size of 0.0001 NIS.

#	Ticker	Bond Name	Bond Type	Median Daily Volume	Average Spread (%)
1	TB810	Makam	Nominal	11,348,764	0.55
2	TB910	Makam	Nominal	6,160,309	0.04
3	TB1010	Makam	Nominal	7,225,325	0.03
4	TB1110	Makam	Nominal	9,165,360	0.05
5	TB1210	Makam	Nominal	4,422,852	0.09
6	TB111	Makam	Nominal	9,329,499	0.05
7	TB211	Makam	Nominal	3,565,230	0.08
8	TB311	Makam	Nominal	3,427,885	0.09
9	TB411	Makam	Nominal	6,829,468	0.05
10	TB511	Makam	Nominal	3,777,397	0.10
11	GN2250	Gilon	Nominal	776,864	0.17
12	GN2251	Gilon	Nominal	545,998	0.35
13	GN2270	Gilon	Nominal	4,432,087	0.10
14	NGN2301	Gilon	Nominal	14,718,775	0.08
15	NGN2302	Gilon	Nominal	11,095,042	0.13
16	SH2633	Shahar	Nominal	3,180,348	0.07
17	SH2634	Shahar	Nominal	3,039,223	0.12
18	SH2635	Shahar	Nominal	4,102,391	0.10
19	SH2636	Shahar	Nominal	2,007,076	0.14
20	SH2660	Shahar	Nominal	3,210,500	0.14
21	SH2661	Shahar	Nominal	16,824,409	0.09
22	KF1399	Kfir	Index	150,988	1.16
23	KF1511	Kfir	Index	933,401	0.57
24	GL3870	Galil	Index	1,504,207	0.91
25	SG4256	Sagi	Index	2,520,046	0.46
26	SG4257	Sagi	Index	1,725,618	0.45
27	GL4703	Galil	Index	227,666	1.88
28	GL5419	Galil	Index	73,999	4.78
29	GL5420	Galil	Index	325,234	3.92
30	GL5422	Galil	Index	56,262	4.76
31	GL5424	Galil	Index	65,851	0.94
32	GL5425	Galil	Index	287,093	1.07
33	GL5426	Galil	Index	548,219	0.68
34	GL5427	Galil	Index	4,407,816	0.27
35	GL5451	Galil	Index	1,444,975	1.37
36	GL5470	Galil	Index	2,052,085	0.42
37	GL5471	Galil	Index	390,702	0.71
38	GB6535	Gilboa	Dollar	412,169	0.48
39	GB6536	Gilboa	Dollar	1,108,129	0.41
40	GB6537	Gilboa	Dollar	1,082,653	0.40
41	GB6538	Gilboa	Dollar	2,011,695	0.58
42	GB6539	Gilboa	Dollar	3,768,712	0.20
43	GB6540	Gilboa	Dollar	11,287,801	0.14
44	GB6541	Gilboa	Dollar	7,968,599	0.52
45	GB6542	Gilboa	Dollar	4,557,639	0.26

Table 2

Summary statistics for time-to-cancellation (hours : minutes : seconds) by order strategy, of non-error limit orders that were cancelled before any transaction took place, in the sample of 32 most liquid stocks traded on TASE, with no changes in the tick-size, and 45 most liquid government bonds, for the sample period May 1, 2000 to July 31, 2000. The number of cancelled orders and the percentage of cancellations for each order category appear in the last two rows of the table.

	Panel A: Stocks			Panel B: Bonds		
	Price-Improving LMT orders	Non-Improving LMT orders	Cancelled LMT Orders	Price-Improving LMT orders	Non-Improving LMT Orders	Cancelled LMT Orders
Mean	0:58:11	2:23:31		2:14:09	2:56:02	
Std	1:43:23	2:40:30		2:04:27	2:09:18	
Min	0:00:03	0:00:03		0:00:04	0:00:03	
Q25	0:02:25	0:07:40		0:20:07	0:46:51	
median	0:11:17	0:58:11		1:29:17	2:46:37	
Q75	0:52:43	4:46:23		4:08:11	5:07:27	
Max	7:14:38	7:14:45		6:04:23	6:04:27	
# of obs	34,950	120,833	155,783	16,281	22,631	38,912
Pct	40%	67%	58%	61%	81%	71%

Table 3

Summary statistics for time-to-first-hit (hours : minutes : seconds) by order strategy, of non-error limit orders, for the sample of 32 most liquid stocks traded on TASE, with no changes in the tick-size, and 45 most liquid government bonds, for the sample period May 1, 2000 to July 31, 2000. Time-to-first-hit is defined to be the minimum of two: the execution-time of the first share for the limit order and the cancellation-time of the limit order (if cancelled). The number and the percentage of orders in each category appear in the last two rows of the table.

	Panel A: Stocks			Panel B: Bonds		
	Price-Improving LMT orders	Non-Improving LMT orders	All LMT Orders	Price-Improving LMT orders	Non-Improving LMT Orders	All LMT Orders
Mean	0:33:16	1:49:32		1:35:49	2:31:39	
Std	1:14:41	2:25:00		1:53:25	2:09:21	
Min	0:00:00	0:00:01		0:00:00	0:00:00	
Q25	0:01:26	0:05:10		0:08:24	0:28:02	
median	0:05:39	0:29:58		0:39:52	1:54:14	
Q75	0:22:45	2:54:12		2:36:32	4:40:56	
Max	7:14:38	7:14:45		6:04:23	6:04:27	
# of obs	87,030	181,070	268,100	26,585	28,052	54,637
Pct	21%	44%	65%	34%	36%	70%

Table 4

Estimates of the parameters of the lifetime equations, for limit orders that were submitted when the spread was two ticks or more, in the sample of 32 most liquid stocks traded on TASE, with no changes in the tick-size, and for the sample of 45 most liquid government bonds, for the sample period May 1, 2000 to July 31, 2000. We assume that the expected lifetimes (minutes) are distributed log-normally and treat cancellations as a naive (uninformative) censoring. ρ_{ju} is the estimate of the correlation between the latent price aggressiveness variable C and the $Lifetime_j$ ($j=1,2$) variables. σ_j is the estimate of the standard deviation of $Lifetime_j$.

Variable	Prediction	Panel A: Sample of Stocks		Panel B: Sample of Bonds	
		Price Improving LMT orders	Non-Improving LMT orders	Price Improving LMT orders	Non-Improving LMT orders
Intercept		17.241 ***	13.128 ***	10.814 ***	18.919 ***
09:45 – 10:45		-0.055	1.046 ***		
10:45 – 11:45		-0.172 ***	0.756 ***	0.319 ***	0.646 ***
11:45 – 12:45		-0.186 ***	0.602 ***	0.028	0.052
12:45 – 13:45		0.054	0.526 ***	0.138	-0.026
13:45 – 14:45		0.066 *	0.507 ***	0.170 *	-0.324 **
14:45 – 15:45		0.020	0.320 ***	0.104	-0.207
FirstDay		-0.090 ***	0.009	0.078	-0.012
MidWeek		-0.165 ***	-0.139 ***	0.129 **	-0.111
SameLMT	+	0.281 ***	0.541 ***	0.409 ***	0.462 ***
OppLMT	-	-0.155 ***	-0.148 ***	-0.106 ***	-0.071 ***
SameMKT	+	0.583 ***	0.231 ***	0.653 ***	0.521 ***
OppMKT	-	-0.303 ***	-0.337 ***	-0.119	-0.218 *
ArrivalRate	-	-0.138 ***	-0.100 ***	-0.029 ***	-0.045 ***
TradeVolume	-	-0.015 **	-0.029 ***	-0.051 ***	-0.054 ***
Imbalance	+	0.175 ***	0.256 ***	-0.155 ***	-0.073
LastTrade	+	1.188 ***	0.573 ***	0.958 ***	0.678 ***
IVolatility	-	-0.161 ***	-0.095	0.422 ***	-0.117
Price		0.140 ***	0.028 ***	-0.644 **	0.952 **
Volume	-	-0.874 ***	-0.553 ***	-0.634 ***	-0.736 ***
Volatility	-	-0.093 ***	0.001	-0.036 **	-0.143 ***
ZVolume	-	-0.135 ***	-0.107 ***	-0.035	-0.096 ***
TA25 Stock		0.275 ***	-0.030		
DualList Stock		-0.027	-0.057 **		
BondType Dollar				0.586 ***	-0.275 ***
BondType Index				0.700 ***	0.154
Sell		-0.247 ***	0.127 ***	-0.261 ***	-0.069
Var-Cov					
ρ_{ju}		-0.733 ***	-0.594 ***	-0.968 ***	-0.969 ***
σ_j		3.345 ***	3.214 ***	3.018 ***	3.441 ***
# of observations		75,021	143,197	20,008	20,189
# of non-censored		45,928 61%	51,828 36%	8,316 42%	4,352 22%
# of censored		29,093 39%	91,369 64%	11,692 58%	15,837 78%

*** P_value < 0.01 ** P_value < 0.05 * P_value < 0.01

Table 5

Estimates of the parameters of the price aggressiveness equation, for orders that were submitted when the spread was two ticks or more, in the sample of 32 most liquid stocks traded on, with no changes in the tick-size, and for the sample of 45 most liquid government bonds, for the sample period May 1, 2000 to July 31, 2000. We present the marginal effects of the explanatory variables on the probability to submit a MKT order. The marginal effects are calculated for each variable, with all other variables at their mean value. The explanatory variables *Lifetime1* and *Lifetime2* are the estimates of the expected log lifetimes, conditional on price-improving (j=1) and non-improving (j=2) limit order strategies.

Variable	Prediction	Panel A: Stocks		Panel B: Bonds	
		Prob of MKT order		Prob of MKT order	
09:45 – 10:45		-0.068	***		
10:45 – 11:45		-0.055	***	-0.076	**
11:45 – 12:45		-0.042	**	-0.055	
12:45 – 13:45		-0.047	**	-0.058	
13:45 – 14:45		-0.047	**	-0.048	
14:45 – 15:45		-0.022		-0.047	
FirstDay		-0.014	***	-0.014	***
MidWeek		-0.002		-0.013	***
Spread	-	-0.073	***	-0.061	***
Tick 100	-	-0.165	***		
Tick 10	-	-0.108	***		
Tick 1	-	-0.052	***		
Lifetime 1	+	0.077	***	0.063	***
Lifetime 2	+	0.024	***	0.013	
IVolatility	+	0.022	***	0.005	
Price		0.091	***	0.116	***
Volume		0.051	***	0.042	***
Volatility	+	0.007	***	0.008	***
ZVolume	+	0.015	***	0.010	***
OrderSize		-0.032	***	-0.019	***
OrderSize*DayEnd	+	0.004	**	-0.001	
Round	-	-0.043	***	-0.028	***
TA25 Stock		-0.035	***		
DualList Stock		0.011	***		
BondType Dollar				0.069	***
BondType Index				0.062	***
Sell		0.002		0.007	*
# of observations		343,265		59,472	
# of (MKT) obs		125,551	36.6%	19,275	32.4%
# of (Improving LMT) obs		74,868	21.8%	20,008	33.6%
# of (Non-Improving LMT) obs		142,846	41.6%	20,189	33.9%

*** P_value < 0.01 ** P_value < 0.05 * P_value < 0.01

Table 6

The results of a cost of waiting calculation, for TEVA stock and TB810 bond orders, that were submitted when the spread was two ticks or more, for the sample period May 1, 2000 to July 31, 2000. We use the model parameter estimates and the sample mean values of the explanatory variables to calculate the expected lifetimes for an average size order and to calculate the probability the trader will use each one of the three order submission strategies. We assume that the cost of submitting a market order (MKT) relative to the cost of submitting a limit order that does not improve on the quoted price (NILMT) is at least equal to the spread. We assume that the cost of submitting a price improving limit order (ILMT) relative to the cost of submitting a limit order that does not improve on the quoted price (NILMT) is at least equal to one tick. We assume that the trader uses a mixed strategy (submits each order type with the appropriate probability) and calculate the expected cost. We repeat the same calculation setting the expected time to execution to different values (zero; average + 1 hour; average + 7 hours).

Panel A: A Representative Stock (TEVA)

Expected Lifetimes		Zero	Sample Average
Order Submission Strategy	Relative Cost of the Strategy (NIS, relative to Non-Improving LMT)	Probabilities	
MKT order	0.4092 (Average Spread)	0.0004	0.5029
Price Improving LMT	0.1000 (One Tick)	0.0025	0.2180
Non-Improving LMT	0.0000	0.9971	0.2791
Lower Bound of the Expected Cost (NIS)		0.0004	0.2276
Average Bid-Ask Spread (NIS)		0.4092	0.4092
Cost as % of the Average Bid-Ask Spread		0.10%	55.61%

Panel B: A Representative Government Bond (TB810)

Expected Lifetimes		Zero	Sample Average
Order Submission Strategy	Relative Cost of the Strategy (NIS, relative to Non-Improving LMT)	Probabilities	
MKT order	0.0590 (Average Spread)	0.0003	0.3323
Price Improving LMT	0.0001 (One Tick)	0.0052	0.3484
Non-Improving LMT	0.0000	0.9945	0.3194
Lower Bound of the Expected Cost (NIS)		0.0000	0.0195
Average Bid-Ask Spread (NIS)		0.0585	0.0585
Cost as % of the Average Bid-Ask Spread		0.03%	33.28%

Table 7

The results of a second stage estimation of the price aggressiveness parameters, for orders that were submitted when the spread was two ticks or more, in the sample of 32 most liquid stocks traded on, with no changes in the tick-size, for the sample period May 1, 2000 to July 31, 2000. The explanatory variables *Lifetime1* and *Lifetime2* are the estimates of the expected log lifetimes, conditional on price-improving (j=1) and non-improving (j=2) limit order strategies. The explanatory variables *ProbExe1* and *ProbExe2* are estimates of the execution probabilities. The log likelihood and the likelihood ratio tests are calculated for the second stage probit procedure.

Variable	Model 1			Model 2			Model 3		
	Coeff	Chi_sq.	P_val	Coeff	Chi_sq.	P_val	Coeff	Chi_sq.	P_val
Spread	-0.196	4398.0	<.0001	0.195	1094.6	<.0001	0.077	101.5	<.0001
Tick 100	-0.490	347.9	<.0001	-0.933	1172.6	<.0001	-0.937	1062.3	<.0001
Tick 10	-0.286	237.1	<.0001	-0.704	1277.7	<.0001	-0.667	1002.1	<.0001
Tick 1	-0.139	134.5	<.0001	-0.384	936.6	<.0001	-0.356	722.9	<.0001
Lifetime 1	0.206	2121.2	<.0001				0.041	43.5	<.0001
Lifetime 2	0.064	217.2	<.0001				0.173	1113.8	<.0001
ProbExe 1				9.277	377.4	<.0001	9.104	360.3	<.0001
ProbExe 2				-5.935	153.5	<.0001	-6.665	191.0	<.0001
# of observations		343,265			343,265			343,265	
Log Likelihood		-355,046			-357,374			-354,132	
Likelihood Ratio Test (model 3 VS 1)								1,828	<.0001
Likelihood Ratio Test (model 3 VS 2)								6,484	<.0001

Table 8: Descriptions and Predicted Signs of the Explanatory Variables (presented for the buy side)

Variables	Description	Predicted Signs		Sources
		Lifetime	Aggress.	
Arrival rate and degree of competition				
SameLMT	Log of monetary depth at the first three levels of bids.	+	NA	Parlour 1998
OppositeLMT	Log of monetary depth at the first three levels of asks.	-	NA	Parlour 1998
SameMKT	Proportion of buy-side market orders to the total buy-side orders (NIS) in the last half hour.	+	NA	FKK, Rosu
OppMKT	Proportion of sell-side market orders to the total sell-side orders (NIS) in the last half hour.	-	NA	FKK, Rosu
ArrivalRate	Log of order volume in half hour prior to order.	-	NA	FKK
Trade Volume	Log of transaction volume in half hour prior to order.	-	NA	FKK
Imbalance	Difference between the value of the same-side and the opposite-side orders in last half hour, scaled by total order value.	+	NA	
Last Trade	Dummy indicating whether the last trade was buyer initiated.	+	NA	
Variables determining the trade-off between the expected time-to-execution and the price aggressiveness				
Spread	Log of quoted bid-ask spread (NIS)	NA	-	FKK, Rosu
Tick*	Fixed effects for four tick size regimes.	NA	-	FKK
Lifetime1	Predicted value of the log of the time to first hit for price-improving limit orders	NA	+	FKK, Rosu
Lifetime2	Predicted value of the log of the time to first hit for non-improving limit orders	NA	+	FKK, Rosu
Control variables for intraday and day-of-the-week patterns				
Daytime	Fixed effects for times of day.	?	+	
Weekday	Fixed effects for the first and last day of trading week	?	+	
IVolatility	Average price range scaled by the range midpoint half hour prior to order.	-	?	
Various control variables				
Price	Average price of the security	?	?	
Volume	Log of the average daily volume three month before sample period	-	+	
ZVolume	Z-score of the daily opening auction volume	-	+	
Volatility	Average daily price range scaled by the midpoint.	-	?	
TA25*	Indicator for inclusion in the TA25 index.	?	?	
DualList*	Indicator for dual listing on Nasdaq or NYSE.	?	?	
OrderSize	Log of order size (NIS)	NA	?	
OrderSize*LastHour	Product of OrderSize and Last Hour indicator	NA	+	Moulton 2005
Sell side order	Indicator of a sell order	?	?	
Round	Indicator of round bid price	NA	-	
BondType**	Fixed effects for dollar, CPI-linked, and nominal bonds	?	?	

*Relevant only for Stocks. **Relevant only for Bonds. ***NIS - New Israeli Shekel

Table A1

Estimates of the parameters of the lifetime equations, for limit orders that were submitted when the spread was two ticks or more, in the sample of 32 most liquid stocks traded on TASE, with no changes in the tick-size, for the sample period May 1, 2000 to July 31, 2000. We assume that the expected lifetimes (minutes) are distributed log-normally and treat cancellations as a naive (uninformative) censoring. ρ_{ju} is the estimate of the correlation between the latent price aggressiveness variable C and the $Lifetime_j$ ($j=1,2$) variables. σ_j is the estimate of the standard deviation of $Lifetime_j$.

Variable	Prediction	Model 1		Model 2	
		Lifetime of a Price Improving LMT order	Lifetime of a Non-Improving LMT order	Lifetime of a Price Improving LMT order	Lifetime of a Non-Improving LMT order
Intercept		19.338***	13.060***	8.946***	-1.610***
09:45 – 10:45		-0.388*	1.268***	0.020	1.238***
10:45 – 11:45		-0.500**	0.979***	-0.040	1.026***
11:45 – 12:45		-0.496**	0.824***	-0.153	0.801**
12:45 – 13:45		-0.229	0.754***	-0.052	0.631*
13:45 – 14:45		-0.211	0.735***	-0.077	0.593*
14:45 – 15:45		-0.233	0.553**	-0.325	0.233
FirstDay		-0.120***	0.006	0.191***	0.274***
MidWeek		-0.190***	-0.141***	0.080***	0.086***
SameLMT	+	0.343***	0.546***	-0.214***	0.092***
OppLMT	-	-0.156***	-0.146***	-0.185***	-0.184***
SameMKT	+	0.678***	0.237***	-0.512***	-0.738***
OppMKT	-	-0.362***	-0.332***	0.338***	0.253***
ArrivalRate	-	-0.146***	-0.099***	0.035***	0.065***
TradeVolume	-	-0.009*	-0.030***	-0.031***	-0.043***
Imbalance	+	0.181***	0.257***	0.126***	0.210***
LastTrade	+	1.331***	0.578***	-0.316***	-0.740***
IVolatility	-	-0.181***	-0.096***	-0.229***	-0.202***
Price		0.160***	0.031***	-1.199***	-1.315***
Volume	-	-0.902***	-0.540***	-0.261***	0.031
Volatility	-	-0.123***	0.002	-0.138***	-0.031**
ZVolume	-	-0.127***	-0.107***	-0.130***	-0.118***
TA25 Stock		0.293***	-0.033	0.513***	0.181***
DualList Stock		-0.009	-0.060***	-0.164***	-0.156***
Sell		-0.246***	0.131***	-0.056**	0.267***
OrderSize	-	-0.181***	-0.039***	0.333***	0.382***
OrderSize*DayEnd		-0.017	0.021	-0.098***	-0.071**
Spread				1.273***	1.277***
# of observations		75,021	143,197	75,021	143,197
# of non-censored		45,928 61%	51,828 36%	45,928 61%	51,828 36%
# of censored		29,093 39%	91,369 64%	29,093 39%	91,369 64%
Var-Cov					
ρ_{ju}		-0.795***	-0.605***	-0.774***	-0.900***
σ_j		3.715***	3.227***	3.559***	4.862***

*** P_value < 0.01 ** P_value < 0.05 * P_value < 0.01

Table A2

The results of a second stage estimation of the price aggressiveness parameters, for orders that were submitted when the spread was two ticks or more, in the sample of 32 most liquid stocks traded on, with no changes in the tick-size, for the sample period May 1, 2000 to July 31, 2000. The explanatory variables *Lifetime1* and *Lifetime2* are the estimates of the expected log lifetimes, conditional on price-improving (j=1) and non-improving (j=2) limit order strategies. The log likelihood is calculated for the second stage probit procedure.

Variable	Prediction	Model 1			Model 2		
		Price aggressiveness			Price aggressiveness		
		Coeff	Chi_sq.	P_val	Coeff	Chi_sq.	P_val
Intercept1		-2.550	899.8	<.0001	-4.953	438.9	<.0001
Intercept2		0.578	91271.1	<.0001	0.580	91297.6	<.0001
09:45 – 10:45		-0.187	11.9	0.0005	-0.088	1.3	0.2483
10:45 – 11:45		-0.149	7.7	0.0055	-0.056	0.7	0.4102
11:45 – 12:45		-0.114	4.5	0.0346	-0.033	0.3	0.6025
12:45 – 13:45		-0.127	5.6	0.0179	-0.098	2.8	0.0948
13:45 – 14:45		-0.127	5.6	0.0184	-0.099	2.9	0.0897
14:45 – 15:45		-0.058	1.2	0.2818	-0.040	0.5	0.4723
FirstDay		-0.037	33.4	<.0001	-0.020	7.9	0.0049
MidWeek		-0.006	1.4	0.2461	0.014	5.9	0.0149
SameLMT	+				-0.002	0.0	0.9158
OppLMT	-				0.069	382.7	<.0001
ArrivalRate	-				0.022	71.3	<.0001
TradeVolume	-				0.008	19.7	<.0001
Imbalance	+				-0.057	41.8	<.0001
LastTrade	+				-0.164	113.3	<.0001
Spread	-	-0.196	4398.0	<.0001	-0.195	4292.3	<.0001
Tick 100	-	-0.490	347.9	<.0001	-0.623	551.3	<.0001
Tick 10	-	-0.286	237.1	<.0001	-0.370	390.2	<.0001
Tick 1	-	-0.139	134.5	<.0001	-0.184	232.6	<.0001
Lifetime 1	+	0.206	2121.2	<.0001	0.391	164.3	<.0001
Lifetime 2	+	0.064	217.2	<.0001	0.001	0.0	0.9888
IVolatility		0.057	452.5	<.0001	0.072	443.1	<.0001
Price		0.243	3332.5	<.0001	0.234	2035.5	<.0001
Volume		0.137	997.5	<.0001	0.227	467.4	<.0001
Volatility		0.018	42.7	<.0001	0.043	114.8	<.0001
ZVolume		0.040	532.8	<.0001	0.053	405.3	<.0001
OrderSize		-0.086	1486.0	<.0001	-0.093	1717.2	<.0001
OrderSize*DayEnd		0.010	4.4	0.035	0.010	4.2	0.0410
Round		-0.115	755.5	<.0001	-0.119	799.5	<.0001
TA25 Stock		-0.093	289.7	<.0001	-0.157	201.8	<.0001
DualList Stock		0.031	34.5	<.0001	0.029	25.9	<.0001
Sell		0.004	0.8	0.3581	0.065	21.5	<.0001
# of observations			343,265			343,265	
Log Likelihood			-355,046			-354,185	